

The Derivative: Instantaneous Rate of Change

David Puelz

The University of Austin

Outline

1. **Review: Average Rate of Change** – The difference quotient (10 min)
2. **The Limit Process** – From average to instantaneous (20 min)
3. **Definition of the Derivative** – Formal statement and notation (15 min)
4. **Computing Derivatives** – Worked examples (25 min)
5. **Summary** – Key formulas and looking ahead (5 min)

1 Review: Average Rate of Change

1.1 The Difference Quotient

Recall from last lecture: For a function $f(t)$, the **average rate of change** over an interval from t to $t + h$ is:

$$\frac{f(t+h) - f(t)}{h}$$

This is called the **difference quotient**.

Geometric interpretation: The difference quotient is the **slope of the secant line** connecting the points $(t, f(t))$ and $(t+h, f(t+h))$ on the graph of f .

1.2 Example: Position and Velocity

Setup: A car's position along a highway is given by $x(t) = t^2$ miles, where t is in hours.

Average velocity from $t = 1$ to $t = 3$:

$$\frac{x(3) - x(1)}{3 - 1} = \frac{9 - 1}{2} = \frac{8}{2} = 4 \text{ mph}$$

Average velocity from $t = 2$ to $t = 3$:

$$\frac{x(3) - x(2)}{3 - 2} = \frac{9 - 4}{1} = 5 \text{ mph}$$

Observation: Different intervals give different average velocities. But what is the velocity at *exactly* $t = 2$?

2 The Limit Process

2.1 Shrinking the Interval

The key idea: To find the **instantaneous** rate of change at $t = 2$, we compute average rates over smaller and smaller intervals.

General average rate starting at $t = 2$:

$$\frac{x(2+h) - x(2)}{h} = \frac{(2+h)^2 - 4}{h}$$

Simplifying:

$$\begin{aligned} \frac{(2+h)^2 - 4}{h} &= \frac{4 + 4h + h^2 - 4}{h} \\ &= \frac{4h + h^2}{h} \\ &= \frac{h(4+h)}{h} \\ &= 4 + h \quad (\text{for } h \neq 0) \end{aligned}$$

2.2 Taking the Limit

Computing average rates for smaller h :

Interval size h	Average rate $= 4 + h$
1	5
0.5	4.5
0.1	4.1
0.01	4.01
0.001	4.001
0.0001	4.0001

The pattern: As $h \rightarrow 0$, the average rate approaches 4.

We write:

$$\lim_{h \rightarrow 0} (4 + h) = 4$$

Conclusion: The instantaneous velocity at $t = 2$ is exactly 4 mph.

2.3 The “Evanescent h ”

Historical note: The philosopher George Berkeley (1685–1753) criticized this process, calling h an “evanescent increment”—something that vanishes like vapor.

Berkeley's objection: How can we divide by h and then set $h = 0$? Isn't that dividing by zero?

The resolution: We never actually set $h = 0$. We examine what happens as h approaches 0. The expression $4 + h$ is defined for all $h \neq 0$, and it approaches 4 as h gets arbitrarily small.

The limit concept: This is the foundation of calculus. We can make the average rate as close to 4 as we like by choosing h sufficiently small.

3 Definition of the Derivative

3.1 Formal Definition

Definition: The **derivative** of a function f at a point t is:

$$f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$$

provided this limit exists.

The derivative $f'(t)$:

- Is the **instantaneous rate of change** of f at t
- Is the **slope of the tangent line** to the graph of f at the point $(t, f(t))$
- Is itself a **function** of t

3.2 Notation

Several notations are used for the derivative:

Notation	Read as
$f'(t)$	" f prime of t "
$\frac{df}{dt}$	" df by dt " or "the derivative of f with respect to t "
$\frac{d}{dt}f(t)$	" d by dt of f of t "
$\mathcal{D}_t f(t)$	" D sub t of f "
$\dot{f}(t)$	" f dot" (mainly in physics, for time derivatives)

All equivalent: These all mean the same thing—the instantaneous rate of change of f with respect to t .

3.3 The Derivative as a Function

Key insight: The derivative takes a function and produces a *new function*.

Example: If $f(t) = t^2$, then $f'(t) = 2t$.

- The original function f tells us the position at any time t
- The derivative f' tells us the velocity at any time t

We can evaluate the derivative at specific points:

- $f'(1) = 2(1) = 2$
- $f'(2) = 2(2) = 4$
- $f'(3) = 2(3) = 6$

4 Computing Derivatives

4.1 Example 1: $f(t) = t^2$

Step 1: Write the difference quotient.

$$\frac{f(t+h) - f(t)}{h} = \frac{(t+h)^2 - t^2}{h}$$

Step 2: Expand and simplify.

$$\begin{aligned} \frac{(t+h)^2 - t^2}{h} &= \frac{t^2 + 2th + h^2 - t^2}{h} \\ &= \frac{2th + h^2}{h} \\ &= \frac{h(2t + h)}{h} \\ &= 2t + h \end{aligned}$$

Step 3: Take the limit as $h \rightarrow 0$.

$$f'(t) = \lim_{h \rightarrow 0} (2t + h) = 2t$$

Result: $\boxed{\frac{d}{dt}(t^2) = 2t}$

4.2 Example 2: $f(t) = t^3$

Step 1: Write the difference quotient.

$$\frac{f(t+h) - f(t)}{h} = \frac{(t+h)^3 - t^3}{h}$$

Step 2: Expand $(t + h)^3$.

$$(t + h)^3 = t^3 + 3t^2h + 3th^2 + h^3$$

Step 3: Simplify the difference quotient.

$$\begin{aligned} \frac{(t + h)^3 - t^3}{h} &= \frac{t^3 + 3t^2h + 3th^2 + h^3 - t^3}{h} \\ &= \frac{3t^2h + 3th^2 + h^3}{h} \\ &= \frac{h(3t^2 + 3th + h^2)}{h} \\ &= 3t^2 + 3th + h^2 \end{aligned}$$

Step 4: Take the limit as $h \rightarrow 0$.

$$f'(t) = \lim_{h \rightarrow 0} (3t^2 + 3th + h^2) = 3t^2$$

Result: $\boxed{\frac{d}{dt}(t^3) = 3t^2}$

4.3 Example 3: $f(t) = 5t + 3$ (Linear Function)

Step 1: Write the difference quotient.

$$\frac{f(t + h) - f(t)}{h} = \frac{[5(t + h) + 3] - [5t + 3]}{h}$$

Step 2: Simplify.

$$\frac{5t + 5h + 3 - 5t - 3}{h} = \frac{5h}{h} = 5$$

Step 3: Take the limit (it's already constant!).

$$f'(t) = \lim_{h \rightarrow 0} 5 = 5$$

Result: $\boxed{\frac{d}{dt}(5t + 3) = 5}$

Interpretation: The derivative of a linear function is its slope. This makes sense—a line has constant steepness everywhere.

4.4 Example 4: $f(t) = c$ (Constant Function)

Step 1: Write the difference quotient.

$$\frac{f(t + h) - f(t)}{h} = \frac{c - c}{h} = \frac{0}{h} = 0$$

Step 2: Take the limit.

$$f'(t) = \lim_{h \rightarrow 0} 0 = 0$$

Result: $\boxed{\frac{d}{dt}(c) = 0}$

Interpretation: A constant function doesn't change, so its rate of change is zero.

4.5 Example 5: $f(t) = \frac{1}{t}$

Step 1: Write the difference quotient.

$$\frac{f(t+h) - f(t)}{h} = \frac{\frac{1}{t+h} - \frac{1}{t}}{h}$$

Step 2: Combine fractions in the numerator.

$$\frac{1}{t+h} - \frac{1}{t} = \frac{t - (t+h)}{t(t+h)} = \frac{-h}{t(t+h)}$$

Step 3: Divide by h .

$$\frac{1}{h} \cdot \frac{-h}{t(t+h)} = \frac{-1}{t(t+h)}$$

Step 4: Take the limit as $h \rightarrow 0$.

$$f'(t) = \lim_{h \rightarrow 0} \frac{-1}{t(t+h)} = \frac{-1}{t \cdot t} = -\frac{1}{t^2}$$

Result: $\boxed{\frac{d}{dt} \left(\frac{1}{t} \right) = -\frac{1}{t^2}}$

4.6 Emerging Pattern: The Power Rule

Let's collect our results:

$f(t)$	$f'(t)$
t^2	$2t^1 = 2t$
t^3	$3t^2$
$t^1 = t$	$1t^0 = 1$
$t^0 = 1$	0
$t^{-1} = \frac{1}{t}$	$-1 \cdot t^{-2} = -\frac{1}{t^2}$

The pattern: For $f(t) = t^n$:

$$f'(t) = n \cdot t^{n-1}$$

This is the **Power Rule**—one of the most important derivative formulas.

5 Summary

5.1 Key Definitions

1. **Difference quotient:** $\frac{f(t+h) - f(t)}{h}$ (average rate of change)
2. **Derivative:** $f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$ (instantaneous rate of change)
3. **Power Rule:** $\frac{d}{dt}(t^n) = n \cdot t^{n-1}$

5.2 Key Results

Function	Derivative
c (constant)	0
t	1
t^2	$2t$
t^3	$3t^2$
t^n	nt^{n-1}
$\frac{1}{t} = t^{-1}$	$-\frac{1}{t^2} = -t^{-2}$

5.3 Looking Ahead

Next lecture: We'll explore derivative notation in depth (especially the $\mathcal{D}_t$ operator) and see how derivatives apply to real-world problems in physics, economics, and biology.

5.4 Practice Exercises

Exercise 1: Use the limit definition to find the derivative of $f(t) = t^2 + 3t$.

Exercise 2: Use the limit definition to find the derivative of $f(t) = 4t^2 - 2t + 7$.

Exercise 3: Verify the power rule for $f(t) = t^4$ by computing the derivative using the limit definition. (You'll need to expand $(t+h)^4$.)

Exercise 4: If $x(t) = t^3 - 6t$ represents position:

- (a) Find the velocity function $v(t) = x'(t)$.
- (b) At what time(s) is the velocity zero?
- (c) What is the velocity at $t = 2$?

Exercise 5: Use the limit definition to find the derivative of $f(t) = \sqrt{t}$. (Hint: Rationalize by multiplying by $\frac{\sqrt{t+h}+\sqrt{t}}{\sqrt{t+h}+\sqrt{t}}$.)

Worked Solutions

Exercise 1 (Solution): We want $f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$ for $f(t) = t^2 + 3t$.

Step 1. Compute $f(t+h) = (t+h)^2 + 3(t+h) = t^2 + 2th + h^2 + 3t + 3h$.

Step 2. So $f(t+h) - f(t) = (t^2 + 2th + h^2 + 3t + 3h) - (t^2 + 3t) = 2th + h^2 + 3h = h(2t + h + 3)$.

Step 3. The difference quotient is $\frac{h(2t+h+3)}{h} = 2t + h + 3$ (for $h \neq 0$).

Step 4. As $h \rightarrow 0$, we get $f'(t) = 2t + 3$. $\square$

Exercise 2 (Solution): For $f(t) = 4t^2 - 2t + 7$, we have $f(t+h) = 4(t+h)^2 - 2(t+h) + 7 = 4t^2 + 8th + 4h^2 - 2t - 2h + 7$. So $f(t+h) - f(t) = 8th + 4h^2 - 2h = h(8t + 4h - 2)$. The difference quotient is $8t + 4h - 2 \rightarrow 8t - 2$ as $h \rightarrow 0$. Thus $f'(t) = 8t - 2$. $\square$

Exercise 3 (Solution): For $f(t) = t^4$, expand $(t+h)^4 = t^4 + 4t^3h + 6t^2h^2 + 4th^3 + h^4$. So $f(t+h) - f(t) = 4t^3h + 6t^2h^2 + 4th^3 + h^4 = h(4t^3 + 6t^2h + 4th^2 + h^3)$. The difference quotient is $4t^3 + 6t^2h + 4th^2 + h^3 \rightarrow 4t^3$ as $h \rightarrow 0$. So $f'(t) = 4t^3$, which matches the power rule $\frac{d}{dt}(t^4) = 4t^3$. $\square$

Exercise 4 (Solution): Given $x(t) = t^3 - 6t$.

(a) $v(t) = x'(t) = 3t^2 - 6$.

(b) Velocity is zero when $3t^2 - 6 = 0$, i.e., $t^2 = 2$, so $t = \sqrt{2}$ or $t = -\sqrt{2}$.

(c) $v(2) = 3(4) - 6 = 12 - 6 = 6$.

$\square$

Exercise 5 (Solution): For $f(t) = \sqrt{t}$ ($t > 0$), the difference quotient is $\frac{\sqrt{t+h} - \sqrt{t}}{h}$. Multiply numerator and denominator by $\sqrt{t+h} + \sqrt{t}$:

$$\frac{\sqrt{t+h} - \sqrt{t}}{h} = \frac{(t+h) - t}{h(\sqrt{t+h} + \sqrt{t})} = \frac{1}{\sqrt{t+h} + \sqrt{t}}.$$

As $h \rightarrow 0$, the denominator $\sqrt{t+h} + \sqrt{t} \rightarrow 2\sqrt{t}$. So $f'(t) = \frac{1}{2\sqrt{t}} = \frac{1}{2}t^{-1/2}$. $\square$